Claude Code

Task	Command	Notes
Install	`curl -fsSL https://claude.ai/install.sh | bash`	native installer (auto-updates)
Install (Homebrew)	`brew install --cask claude-code`	alternative
Update	`claude update`	or auto-updates on launch
Diagnose	`claude doctor`	check installation health
Auth	`claude auth login`	`--sso` for SSO, `--console` for API key
Version	`claude -v`	
Editors	`SPC ac` nvim, `C-c C-'` Emacs	see the editor cards

CLI Flags — Essentials

`-p "query"`	Print mode (non-interactive)
`-c`	Continue last session
`-r id`	Resume specific session
`-n "name"`	Name session
`-w name`	Run in git worktree
`--model opus`	Set model
`--fast`	Fast mode

CLI Flags — Permissions

`--permission-mode m`	default / plan / acceptEdits / dontAsk
`--allowedTools "..."`	Auto-approve tools
`--disallowedTools "..."`	Remove tools

CLI Flags — Output

`--output-format text`	Plain text (default)
`--output-format json`	JSON with metadata
`--output-format stream-json`	Streaming JSON
`--json-schema '{...}'`	Structured output

CLI Flags — Advanced

`--effort high`	low / medium / high / max
`--max-turns 5`	Limit agentic turns
`--max-budget-usd 5`	Spending cap
`--system-prompt "..."`	Replace system prompt
`--append-system-prompt`	Add to system prompt
`--mcp-config f.json`	Load MCP servers
`--add-dir ../lib`	Add working dirs
`--debug`	Debug mode
`--verbose`	Verbose logging
`--remote`	Web session on claude.ai

Slash Commands — Session

`/clear`	New conversation
`/resume`	Resume session picker
`/rename` *name*	Rename session
`/compact`	Compress context
`/branch`	Fork conversation
`/copy`	Copy last response
`/export`	Export as text
`/exit`	Quit

Slash Commands — Info

`/help`	Show commands
`/status`	Version, model, account
`/cost`	Token usage
`/context`	Context usage grid
`/doctor`	Diagnose install
`/stats`	Daily usage stats

Slash Commands — Config

`/config`	Settings UI
`/model` *name*	Change model
`/effort` *level*	Change effort
`/fast`	Toggle fast mode
`/vim`	Toggle vim mode
`/theme`	Change theme
`/terminal-setup`	Terminal keybindings
`/permissions`	View/manage rules

Slash Commands — Tools

`/memory`	Edit CLAUDE.md / memory
`/init`	Init project CLAUDE.md
`/plan`	Enter plan mode
`/diff`	Interactive diff viewer
`/mcp`	Manage MCP servers
`/hooks`	View hook configs
`/skills`	List skills
`/agents`	Manage subagents
`/review`	Code review

Keyboard Shortcuts

`Ctrl+C`	Cancel / interrupt
`Ctrl+D`	Exit session
`Ctrl+L`	Clear screen
`Ctrl+R`	Search history
`Ctrl+G`	Open in editor
`Ctrl+O`	Toggle verbose
`Ctrl+T`	Toggle task list
`Ctrl+V`	Paste image
`Shift+Tab`	Cycle permission modes
`Alt+P`	Switch model
`Alt+T`	Toggle thinking
`Esc Esc`	Rewind

Multiline Input

`Option+Enter`	macOS default
`Shift+Enter`	After terminal-setup
`\ + Enter`	Quick escape
`Ctrl+J`	Control sequence

Model Aliases

`sonnet`	Latest Sonnet
`opus`	Latest Opus
`haiku`	Fast / cheap
`default`	Plan default
`opus[1m]`	Opus 1M context
`sonnet[1m]`	Sonnet 1M context

Claude Code — Configuration & Hooks

CLAUDE.md Files

`./CLAUDE.md`	Project (shared via git)
`./.claude/CLAUDE.md`	Alt project location
`~/.claude/CLAUDE.md`	User (all projects)
`.claude/rules/*.md`	Scoped rules
`@path/to/file`	Import syntax

Rules support `paths:` frontmatter to scope to specific files. Keep CLAUDE.md <200 lines.

Settings Files

`~/.claude/settings.json`
User settings

`.claude/settings.json`	Project (shared)

`.claude/settings.local.json`
Project (local, gitignored)

Precedence: user < project < local < CLI args < managed.

Settings Keys

`model`	Default model alias
`effortLevel`	low / medium / high
`autoMemoryEnabled`	Auto-memory on/off

`permissions.defaultMode`
Default perm mode

`permissions.allow`	Auto-approve list
`permissions.deny`	Block list
`hooks`	Hook definitions
`env`	Environment vars
`sandbox.enabled`	Sandbox mode

Permission Rules

`"Bash"`	Entire tool
`"Bash(npm run *)"`	Glob on command
`"Bash(git push *)"`	In deny list
`"Edit(src/**/*.ts)"`	Glob on file path
`"Read(~/.ssh/*)"`	Block sensitive files

`"WebFetch(domain:gh.com)"`
Domain filter

`"mcp__server__*"`	MCP tool pattern

Eval order: deny → ask → allow.

Permission Modes

`default`	Prompt on first use
`plan`	Read-only analysis
`acceptEdits`	Auto-accept edits
`dontAsk`	Deny unless allowed
`bypassPermissions`	Skip all prompts

Hook Events

`PreToolUse`	Before tool (can block)
`PostToolUse`	After tool success
`UserPromptSubmit`	User sends prompt
`SessionStart`	Session begins
`SessionEnd`	Session ends
`Stop`	Claude finishes
`Notification`	Needs attention
`SubagentStart/Stop`	Subagent lifecycle

`PreCompact/PostCompact`
Context compaction

Hook Types

`command`	Run shell command
`http`	POST to endpoint
`prompt`	Single-turn LLM
`agent`	Multi-turn subagent

Exit codes: 0 = allow, 2 = block.

Hook Example

```
"hooks": {
  "PostToolUse": [{
    "matcher": "Edit|Write",
    "hooks": [{
      "type": "command",
      "command": "prettier --write
        \"$TOOL_INPUT.file_path\""
    }]
  }]
}
```

MCP Configuration

`~/.claude/mcp.json`	User-scope servers
`.claude/.mcp.json`	Project-scope
`--mcp-config f.json`	Session-scope
`claude mcp add` *n*	Add server (CLI)
`claude mcp remove` *n*	Remove server
`claude mcp list`	List servers

MCP Server Config

```
{
  "mcpServers": {
    "name": {
      "command": "/path/to/server",
      "args": ["arg1"],
      "env": {"KEY": "value"}
    }
  }
}
```

Memory System

`~/.claude/projects/*/memory/`
Auto-memory dir

`MEMORY.md`	Index (auto-loaded)
`*.md`	Topic files
`/memory`	Manage memory

Environment Variables

`ANTHROPIC_API_KEY`	API key auth
`ANTHROPIC_MODEL`	Default model

`CLAUDE_CODE_EFFORT_LEVEL`
Effort level

`MAX_THINKING_TOKENS`	Thinking budget
`DISABLE_AUTOUPDATER`	Stop auto-updates

Headless Patterns

```
# JSON output
claude -p "query" \
  --output-format json | jq '.result'

# Auto-approve tools
claude -p "query" \
  --allowedTools "Bash,Read,Edit"

# Pipe input
cat f.txt | claude -p "analyze"

# Continue session
claude -p "follow-up" \
  --resume $SESSION_ID
```